Supplementary Material for MRV: Evidence-Constrained Ordering for Byzantine Fault-Tolerant DAG Ledgers

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This supplementary material provides supporting proofs and implementation correspondence for MRV's structural precedence guarantees. It also documents the baseline integrations, visibility-edge ablations, Byzantine withholding executions, and online payoff audits. Terminology and execution-slice definitions follow the main paper; a named *main-paper theorem* denotes the corresponding result in the accompanying manuscript.

I. Supporting Proofs

This section expands the main paper's ordering, acyclicity, structural-validity, and sealing arguments and proves comparator correctness. Section II connects these guarantees to the prototype's exporter and visibility maintenance.

A. Comparator Correctness

Lemma S1 (Comparator Correctness): For causally incomparable $A, B \in S$, the main paper's PAIREDGE algorithm returns $\mathsf{Edge}(A \to B)$ if and only if $\mathsf{Edge}_S(A, B)$ holds.

Proof. The eligibility check returns INELIGIBLE exactly when either target is ineligible. Otherwise, the complete-window scan sets pos_{AB} precisely when $\mathsf{Pos}_S(A, B)$ holds. Since $\mathcal{W}(A, B) = \mathcal{W}(B, A)$ and $\Delta_S(B, A, t) = -\Delta_S(A, B, t)$, pos_{BA} similarly represents $\mathsf{Pos}_S(B, A)$. The return branches therefore implement Edge_S and its symmetric case, detecting both-direction crossings as CONFLICT and the absence of both crossings as NOSIGNAL.

B. Deterministic Per-Slice Ordering and No Rollback

Proof of the main-paper Deterministic Ordering and No Rollback theorem. Under the exporter contract, all correct replicas observe the same execution-slice and committed-prefix sequences. These fix S, its export event $o(S)$, the least qualifying index $k_S^\star$, and the snapshot $P_{k_S^\star}$. The common snapshot contains identical canonical creator-round vertices and authenticated ancestry. It therefore determines the same visibility counts, eligibility flags, pair outcomes, and causal edges, yielding identical graphs G_S. The unique SCC partition and fixed total keys $\kappa(C)$ and $\kappa(X)$ determine identical component and vertex orders. Sealed orders depend only on $P_{k_S^\star}$ and the common parameters; later prefixes cannot revise them.

C. Ordering Edges and Causal Refinement

Proof of the main-paper Structural and Global Causal Refinement theorem. Every inter-SCC edge of H_S follows the condensation DAG, while intra-SCC edges are causal and strictly increase round. Thus H_S is acyclic. MRV orders SCCs topologically and uses a causal topological order within each SCC, preserving every edge of H_S and its transitive closure. Each visibility edge either joins SCCs in topological order or remains within one SCC, giving $\beta_S(A) \le \beta_S(B)$. Since $E_S^c \subseteq E(H_S)$, causal reachability is preserved. Any surviving visibility edge joins causally incomparable vertices and is therefore absent from $(E_S^c)^+$, establishing strict refinement.

For global causality, if $B \in \mathsf{Anc}(A)$ and both vertices belong to one slice, the hard causal edge $B \to A$ enforces their order. Otherwise, causal delivery gives $\sigma(B) < \sigma(A)$, and concatenating sealed slices in exporter order places B before A.

D. Quorum-Induced Acyclicity

Proof of the main-paper Quorum-Induced Acyclicity and Precedence theorem. Fix the ancestry-closed snapshot $P = P_{k_S^\star}$ and let $\mathcal{V}_t$ denote its round-t vertex set. For a non-genesis target $X \in S$, write $V_X(t)$ for the creators whose vertices in $\mathcal{V}_t$ contain X in their reflexive ancestry, and $c_X(t) = |V_X(t)|$. Certificate uniqueness ensures $c_X(t) = C_X^{(k_S^\star)}(t)$.

First, suppose $c_X(t) \ge \theta$. Since X is non-genesis, $t \ge r(X) \ge 1$, so the preceding-round parent rule applies to every vertex in $\mathcal{V}_{t+1}$. Its parent-creator set intersects $V_X(t)$ because $q + \theta > n$. Ancestry closure places the referenced parent in $\mathcal{V}_t$, and actual creator-round certificate uniqueness identifies it with the vertex counted in $V_X(t)$. Thus every vertex in $\mathcal{V}_{t+1}$ contains X. For each subsequent round, every vertex has a preceding-round parent, which already contains X; here $q > 0$ follows from $c_X(t) \ge \theta$, $c_X(t) \le n$, and $q + \theta > n$. Induction therefore gives $c_X(u) = |\mathcal{V}_u|$ for every $u > t$. An empty round

causes no difficulty: ancestry closure and the preceding-round parent rule also preclude a later vertex whose parent chain crosses that empty round.

Define $\tau_X = \min\{t : c_X(t) \geq \theta\}$, with $\min \varnothing = \infty$, and let $a_X = c_X(\tau_X)$ when $\tau_X < \infty$, or $a_X = 0$ otherwise. For a visibility edge $X \to Y$ witnessed at t, $c_X(t) - c_Y(t) \geq \theta$ implies $\tau_X \leq t < \infty$. If $\tau_Y < t$, propagation gives $c_Y(t) = |\mathcal{V}_t| \geq c_X(t)$, a contradiction. Hence $\tau_X \leq t \leq \tau_Y$. When $\tau_X = \tau_Y$, both equal t and $a_X - a_Y \geq \theta > 0$. For a hard causal edge $X \to Y$, ancestry gives $V_Y(t) \subseteq V_X(t)$ at every round. Consequently, $\tau_X \leq \tau_Y$. When $\tau_X = \tau_Y$, $a_X \geq a_Y$; if both values are equal, causality requires $r(X) < r(Y)$.

Suppose G_S contains a directed cycle. Since τ is nondecreasing along every edge, it must be constant around the cycle. With τ fixed, a is nonincreasing along every edge and strictly decreases along each visibility edge. Thus a must also be constant, and the cycle cannot contain any visibility edge. All remaining edges are causal and strictly increase round, which is impossible on a cycle. Therefore G_S is acyclic, every SCC is a singleton, and every visibility edge is preserved in every topological completion.

E. Conditional Structural Validity

Proof of the main-paper Conditional Structural Validity theorem. Correct-only witnesses establish eligibility. At the direction witness t, $\Delta_S(A, B, t) \geq \Delta_{\mathsf{cor}}(A, B, t) - f \geq \theta$. If a reverse crossing occurs at $u < t$, then $C_B^{(k_S^\star)}(u) \geq \theta$. The propagation argument in Section I-D implies $\Delta_S(A, B, t) \leq 0$, contradicting the crossing at t. If $u > t$, propagation from $C_A^{(k_S^\star)}(t) \geq \theta$ instead gives $\Delta_S(A, B, u) \geq 0$, excluding a reverse crossing. The case $u = t$ is impossible since $\theta > 0$. Thus $\mathsf{Edge}_S(A, B)$ holds, and the main-paper Quorum-Induced Acyclicity and Precedence theorem makes it binding in every topological completion. Compatible inputs preserve both eligibility witnesses and the correct margin, so the conclusion holds throughout the fixed-facts family.

F. Compatible-Input Example for Conditional Structural Validity

Two compatible model inputs satisfy the quorum premises with $n = 4$, $f = 1$, $q = q_{\text{vis}} = 3$, $\theta = 2$, and $W = 4$. Both targets have correct-only eligibility witnesses, and both inputs have $\Delta_{\mathsf{cor}} = [3, 0, 0, 0]$ over the pair window. Changing one faulty round-4 vertex from B-only to both-target ancestry changes the aggregate margins from $[2, 0, 0, 0]$ to $[3, 0, 0, 0]$. Both constraint graphs force $A \prec_S B$, whereas causal-only completion admits B first. The artifact provides the compatible model inputs, recomputed header hashes, and a verifier.

G. Conditional Sealing

Proof of the main-paper Conditional Sealing theorem. Since S is finite and nonempty and W is finite, $T_r(S) = \max_{X \in S} r(X) + W$ is finite. If the exporter exposes a prefix P_k with $k \geq o(S)$ and $\rho(k) \geq T_r(S)$, the set defining $k_S^\star$ is nonempty and has a least element. For every $X \in S$, the eligibility interval ends at $r(X)+W \leq T_r(S)$. For every compared pair $A, B \in S$, its window ends at $s(A, B)+W \leq T_r(S)$. All queried rounds are therefore evaluated on the fixed snapshot $P_{k_S^\star}$. An absent creator-round vertex contributes zero and does not require waiting for a later prefix. Visibility extraction, eligibility checks, pair comparisons, graph construction, SCC computation, and keyed topological sorting operate on finite inputs. MRV therefore seals S after processing $P_{k_S^\star}$ and completing finite local computation.

II. Implementation Correspondence and Complexity

A. Prototype Correspondence to the Exporter Contract

Ancestry and execution closure. The primary synchronizer establishes parent availability before storing a certificate and passing it to consensus. For each committed leader, the exporter traverses authenticated parents, retrieving certificates from the archive when necessary. Traversal stops at genesis, digests visited in the current traversal, or previously delivered digests; missing required archive entries cause an explicit failure. Inductively, previously delivered vertices already have their execution ancestors in the delivered prefix. The exporter collects all remaining execution ancestors and orders each slice by round, delivering ancestors no later than their descendants, realizing $V_k = V_0 \cup \bigcup_{o(S_i) \leq k} S_i$, with genesis set V_0 and authenticated parent edges E_k.

Exact delivery and first appearance. Persistent digest markers survive cache collection, while the archive retains certificates needed for ancestor retrieval. Each execution vertex is delivered exactly once, and archive-only certificates enter the formal prefix only upon export. Processing newly exported certificates in nondecreasing round order registers ancestor targets before their descendants; causal delivery ensures the same property across events.

Incremental visibility and active-state collection. Every path from a contributor to an active target X contains only vertices of round at least $r(X)$. Retaining certificates at or above the minimum active-target round therefore preserves these paths. If a newly exported target lowers this floor, no previously exported descendant can contain it, since causal delivery would already have exported the target; its relevant paths arrive with the new event. Registration includes self-observation. Each exported canonical creator-round vertex contributes once, with creator sets deduplicating observations per target and round. Inductively, retained contributions and new ancestry observations yield $C_X^{(k)}(t)$ for each active target. Processing the complete sealing event therefore supplies $\mathbf{C}_S$. Slice-local evidence is reclaimed after sealing; sealed orders and queued members remain until release in slice sequence. Section IV-F reports retained-history costs.

Certificate uniqueness and GC. A correct voter retains its creator-round voting record until its monotonically advancing GC boundary passes that round and rejects conflicting headers while the record remains. Before every vote, including on a waiter return, it checks that the header round is at least the current boundary. Thus, it signs at most one header per creator-round. When any two certificate quorums intersect in

more than f voters, their common correct voter excludes conflicting certificates, establishing actual certificate uniqueness. The main-paper Quorum-Induced Acyclicity and Precedence theorem additionally requires the actual distinct-creator parent threshold q and margin θ to satisfy $q + \theta > n$.

B. Prototype Complexity and Retention Accounting

Let $m = |S|$, $L_S = T_r(S) - \min_{X \in S} r(X) + 1$, and $W \geq 1$. The logical visibility table for an unsealed slice has $O(mL_S)$ target-round entries, and the ordering graph has $O(m^2)$ edge-storage space in the worst case. Given constant-time access to count entries, examining all pair windows takes $O(m^2W)$ time. Causal classification makes at most one ancestry query for each unequal-round unordered pair of slice vertices; equal-round pairs require no ancestry query. The cost of one query depends on the retained committed DAG and its ancestry index and is not included in $O(m^2W)$.

For the implementation, every active target has an ordered round map whose entries store deduplicated creator sets. Let $h \geq 1$ be an upper bound on the number of round entries per target at sealing. Creator-set storage for the targets of S is $O(nmh)$ in the worst case. Ordered-map lookups introduce a $\log(1 + h)$ factor, giving $O(m^2W \log(1 + h))$ for direct pairwise comparisons, separate from ancestry classification. Retaining all pair outcomes for measurement adds at most $O(m^2)$ entries. SCC construction and keyed topological sorting require $O(|E(G_S)| + m \log m)$ time.

These are slice-local costs: the complete implementation additionally maintains a shared certificate cache, primary archive, persistent digest delivery markers, and a queue of sealed-but-unreleased outputs. Creator-round observations accumulate until the selected export event has been fully processed; entries above $T_r(S)$ are retained until that slice seals. Once a slice seals, MRV discards its visibility state, eligibility flags, pair outcomes, and graph, but holds its sealed order until all preceding slices have been released. Cached certificates at or above the minimum active-target round and queued sealed members are retained as needed; the archive and persistent delivery markers account for distinct longer-lived histories. The per-slice bounds cover active evidence and ordering state; Section IV-F separately reports process RSS and retained-history measurements.

III. Baseline Implementation Details

A. FairDAG-RL Baseline

Our FairDAG-RL (Rust) baseline derives from Herring's weak-edge-free Narwhal/Tusk reimplementation [1]. It retains serial post-consensus processing, implicit missing-edge resolution, the pairwise dependency rule, SCC processing, and finalization, without adopting Herring's parallel graph construction or FairUpdate mechanism. We use $\gamma = 1$, for which the adaptation requires $n \geq 4f+1$; fault injection models silent crashes. The baseline includes Herring's catch-up repair, which incorporates ordering indicators recorded before a transaction reaches the threshold for dependency-graph insertion. This prevents those indicators from being omitted from subsequent pairwise weight computation.

Our benchmark integration records local receipt order for transactions received directly from clients or through other workers' batches, replaces fixed-capacity dependency storage with dynamically growing storage, and emits per-transaction final-order records. These adaptations preserve the pairwise dependency rule, SCC processing, and finalization rule. Throughput and latency are measured at final-order emission.

B. DoD-Style Baseline

Our DoD-style baseline independently implements an explicit transaction-order graph workflow on Narwhal/Tusk, following the architecture described by DoD [2]. Clients broadcast transactions to replicas; workers build and disseminate local-order graphs, derive a global-order graph from quorum inputs, and commit its digest through Tusk. The worker then resolves graph dependencies and emits the final UID order. This integration supplies a systems-cost reference for explicit transaction-order processing in our benchmark environment.

The implementation includes missing-edge propagation, cross-round support accumulation, and dependency-aware output with processed-transaction feedback. Its local missing-edge store retains the active frontier. The configuration used for the local sample measurements retains cross-round pair support for up to $K = 256$ global-order sequence intervals, with a capacity of 32,768 missing-edge observations, and includes deterministic completion for dependencies remaining unresolved at the retention boundary. These retention settings are engineering choices of the baseline. Ordinary transactions share dependencies when their synthetic keys $tx_id \bmod 100$ match. Latency probes are excluded from dependency-edge construction. Additional sampling preserves ordinary transactions' kind, UID, key, and dependency semantics; Section III-C reports the common-endpoint comparison.

C. DoD Transaction-Sample Validation

We compare ordinary dependency-bearing transactions and probes in one four-validator run at each of 400 and 800 transactions/s, with approximately 60 seconds of input on a single host. Both populations use client submission to the first final-order UID output at the canonical worker after dependency processing. Outputs are deduplicated by UID; quantiles use nearest rank. All sampled transactions complete: 191 ordinary transactions and 4,824 probes at 400 transactions/s, and 431 ordinary transactions and 4,832 probes at 800 transactions/s. Within each run, mean latency differs by less than 3 ms between the two populations; their p50, p95, and p99 latencies differ by at most 43 ms. These observations support closely aligned sampled latency distributions at the two tested loads.

IV. Additional Evaluation Details

A. Frozen Structural Evidence Audit

Across 391 frozen four- and seven-validator executions, we reconstruct 16,083 sealed slices and 121,889 causally

incomparable same-slice vertex pairs. MRV identifies 13,822 qualifying visibility edges and 14,812 retained noncausal precedence relations. All direct visibility edges satisfy the monotonic relationship used in the main paper's acyclicity proof, and the final orders preserve their required precedence. A separate four-replica exporter reconstruction matches all 187 slice memberships and 1,457 delivered digests.

B. Window Sensitivity and Offline Snapshot Replay

The separate local window study uses $n \in \{4, 7\}$ and $W \in \{2, 3, 4, 8\}$, with one native run per configuration, 30 s warmup, 300 s measurement, and 20 s drain. We replay snapshot selection and visibility evaluation over the original 391-run frozen corpus, preserving committed export-event boundaries and slice membership. The full $W = j$ variants, $j \in \{1, 2, 3\}$, jointly use j-round eligibility and comparison windows and select the first committed prefix reaching the corresponding fixed round horizon.

For offline sensitivity analysis, the adaptive variant A4 retains the $W = 4$ eligibility and comparison windows. For each complete committed export event k, define $\tau_X^{(k)} = \min\{t \geq r(X) : C_X^{(k)}(t) \geq \theta\}$, where $\min \varnothing = \infty$. A4 selects the first prefix P_k, $k \geq o(S)$, satisfying either the original $W = 4$ horizon or the gate: $\tau_X^{(k)} < \infty$ for all $X \in S$, and $\rho(k) \geq \max_{X \in S} \tau_X^{(k)} + 1$. The gate uses only the current complete export event. Absent creator-round vertices contribute zero, and the defined evidence windows remain unchanged.

Of the 14,812 baseline noncausal precedence relations, the full $W = 1, 2, 3$ policies retain 0%, 77.0%, and 93.1%, respectively. The $W = 3$ policy advances no snapshots. In contrast, A4 selects earlier committed prefixes for 10,296 of 16,083 slices (64.0%) while retaining 14,317 baseline relations (96.7%). All 55 baseline-enforced withholding-target relations and the recorded resource outcomes across 54 joint executions and 408 sequence opportunities remain unchanged. In this offline replay, earlier snapshot selection retains most observed structural precedence without shortening the eligibility or comparison windows. The evaluated MRV protocol uses the fixed-horizon snapshot rule in the main paper.

C. Evidence Constraints and Completion

We compare completion policies on identical committed snapshots from the original 391-run corpus, containing 14,812 retained noncausal precedence relations. For randomized completion, each slice uses a shared seed and a creator-round key

$$R_S(X) = \mathrm{SHA256}\big(seed_S \| r(X) \| \mathsf{cr}(X)\big),$$

with κ resolving hash collisions. The 32 trial seeds are generated after protocol actions and trace inputs are fixed. This offline analysis isolates the effect of completion keys on the recorded evidence.

Randomized causal-only completion does not guarantee evidence-qualified precedence. Across 32 offline trials, it reverses 6,762–7,153 retained relations, whereas MRV preserves every relation under the same randomized keys, including all 55 enforced withholding-target relations. Direct structural-key sorting (by $(\tau_X, -a_X, \kappa(X))$) also satisfies these constraints. The 55 target relations belong to this frozen-corpus replay; Section IV-D reports the 100 enforced target comparisons from the joint-policy and extension cohorts.

TABLE I
PRESERVATION OF MRV'S 14,812 RETAINED PRECEDENCE RELATIONS ON FROZEN DAG EVIDENCE. RANGES ARE MINIMA AND MAXIMA ACROSS 32 POST-DECISION RANDOMIZED-KEY TRIALS. TARGET RESULTS COVER 55 ENFORCED WITHHOLDING-TARGET RELATIONS.

Completion policy	Reversed	A-first /55
Causal+R	6,762–7,153	8–18
MRV graph+R	0	55
MRV graph+κ	0	55
Full structural key	0	55

Completion sensitivity. Across 151,668 offline parent-selection counterfactual cases preserving H_S, the ordered vertices' κ keys, and correct-creator ancestry, MRV retains the same completion, while direct structural-key sorting changes unconstrained relative orders in 11,174 cases.

In a synthetic SQLite resource-reservation replay, MRV and causal-plus-κ produce different ownership for 95 of 22,892 contended resources under the highest-contention stock-one setting; these differences arise from qualifying visibility constraints. The no-contention control produces no ownership differences.

Independent processing-order audit. A pre-specified study uses 12 all-correct multiprocess runs ($n \in \{4, 7\}$, $W = 4$), crossing zero and fixed heterogeneous link delays with three identity configurations. For each certificate, replica processing order follows the last non-suspended `primary::coreProcessing` log event. Strict-majority references require 3/4 and 4/7 replicas, respectively. All 3,394 reconstructed sealed MRV orders match the orders emitted by the prototype. Among same-round incomparable pairs with binding MRV precedence (29.5–45.5% coverage), MRV agrees with the majority reference in 99.2–100% of pairs per run, versus 63–72% for Tusk, and achieves higher agreement in all 12 runs.

D. Certificate-Withholding Reproduction

Initial and targeted studies. The initial matrix crosses 20 identity configurations and four predefined release conditions, producing 80 executions: 17 same-slice comparisons, 47 cross-slice cases, 13 causal cases, and three unmatched targets. Each role submits a 512-byte target and 47 fillers through normal worker batching. MRV enforces $A \prec B$ in all 17 same-slice comparisons, whereas Tusk and causal-plus-κ place B first. A separate 52-execution R/E/P study evaluates three legal release/parent policies. R and E require an authenticated signal, the certificate for A, and at least one correct round-4 header; R waits for all three correct headers before resuming its proposal, whereas E resumes immediately after the release decision. P follows E's readiness and timing conditions but prefers B-only ancestry among legal, authenticated, ancestry-

complete parent quorums. The study yields 36 same-slice comparisons, 12 cross-slice P cases, one causal reference, and three target anomalies. MRV enforces $A \prec B$ in all 36 same-slice cases, whereas Tusk and causal-plus-κ place B first.

Joint-policy reproduction. The 54-execution study crosses release conditions H0/H1/H3 with parent policies F/P/J across three identity configurations and two repetitions. All release conditions require the authenticated signal and the certificate for A; H0/H1/H3 additionally require zero, one, or three correct round-4 headers. F follows E's fixed parent-selection rule. P prioritizes B-only ancestry, while J prioritizes ancestry containing both targets. All three policies select legal parent quorums under the corresponding H0/H1/H3 release condition. Among 36 same-slice comparisons, MRV enforces $A \prec B$ in 30, whereas Tusk and causal-plus-κ output $B \prec A$ in all 36. F and J each yield 15 enforced A-first cases, two same-slice B-first cases, and one target-mapping anomaly. P yields 16 earlier-slice and two NoSignal B-first outcomes. Under the predefined first-executed-wins resource model, these 30 enforced comparisons assign ownership to A under MRV and to B under both Tusk and causal-plus-κ.

Independent native extension. We extend the controlled single-host loopback study with 132 independently generated native executions: 108 with four validators and 24 with seven validators, each configuration having one active Byzantine creator. The extension uses new identity configurations and existing legal release and parent policies, rather than replaying message schedules.

The four-validator matrix combines six identity configurations, H0/H1/H3 release conditions, F/P/J parent policies, and two repetitions. The seven-validator matrix combines six identity configurations, E/P policies, and two repetitions.

The new executions reproduce the original conditional ordering effect: MRV enforces $A \prec B$ in 70 of 84 predefined same-slice comparisons, or 70 of all 132 executions. Removing visibility constraints admits a causal B-first completion in all 70 enforced cases. Under the predefined first-executed-wins model, these constraints change resource ownership relative to Tusk and causal-plus-κ. The P policy remains able to obtain B-first execution through earlier-slice placement or absent qualifying evidence. In its 48 new executions, 44 are cross-slice and four are NoSignal. These cases lie outside MRV's within-slice precedence guarantee. The artifact retains complete classifications, transaction mappings, certificate checks, and execution records. These counts characterize the tested release and parent policies in controlled native executions.

Availability controls. A separate 48-execution F/P study varies release timing and parent availability across three identity configurations. Among ten incomparable same-slice comparisons, MRV enforces $A \prec B$ in six cases where Tusk places B first; the other four yield NoSignal. Of the remaining 38 executions, A enters an earlier slice in 33, B in four, and one has no usable target mapping.

Visibility-edge ablation. The visibility constraints account for the enforced precedence: removing all visibility edges restores $B \prec A$ under κ in every same-slice comparison-domain execution. Removing the direct $A \to B$ edge admits a reverse topological completion in all 17 initial-matrix, 36 R/E/P, and 30 joint-study edge-bearing graphs. With κ fixed, single-edge deletion yields $B \prec A$ in 12/17, 20/36, and 23/30 cases, respectively.

Round-preserving randomized-key ablation. We replay 4,987 sealed slices from the existing 54 joint and 132 extended executions using 32 post-decision seeds. Both completion variants use the same key

$$R_S(X) = \mathrm{SHA256}\big(seed_S \| r(X) \| \mathsf{cr}(X)\big),$$
$$K_R(X) = \big(r(X), R_S(X), \kappa(X)\big).$$

Among the 100 previously enforced same-slice comparisons (30 joint and 70 extended), B precedes A by one round. Causal-only completion retains B-first under every seed, whereas MRV preserves A-first in all 100 cases.

Across 5,246 retained noncausal precedence relations, causal-only completion reverses 2,553–2,721, including 1,591–1,759 same-round relations. MRV preserves all 5,246 relations under the same keys. These are offline ordering replays over existing native traces, not additional protocol executions.

Commitment-path placement. In 12 targeted P executions, a legal B-only three-parent quorum allows the actor's round-4 certificate to become the committed leader, placing B in slice 2 and A in slice 3. These executions use the prototype's round-robin leader schedule and expose the effect of commitment-path placement. MRV preserves the resulting inter-slice order as required by its exporter contract.

E. Unknown-Value Strategy Sequences

A separately frozen sequence study evaluates 17 online policies across three identity configurations, with two runs per policy/configuration and four predeclared opportunities per run: 102 runs and 408 opportunities. The prearranged high/low value sequence is hidden from the actor until authenticated signals arrive. The policies vary release decisions and legal parent choices: signal-conditioned policies use authenticated value information, whereas prereveal controls make decisions without it. Both have the same legal proposal and parent permissions. Resource payoffs are evaluated offline as $u_B = V\mathbf{1}[B \text{ wins}] - \mathbf{1}[B \text{ is output}]$.

MRV reduces same-trace adversarial payoff in 25 of 408 opportunities (12 under F and 13 under J) and increases it in none. An audit of transaction records, certificates, and cutoff outputs resolves all 408 payoffs. The audited cases include 66 opportunities with zero payoff under both orderers: 57 lack a formed B certificate, and nine have a formed but uncommitted B certificate.

Conditional structural validity audit. Reconstruction from frozen traces verifies the correct-only eligibility witnesses and directional margin required by the main-paper Conditional Structural Validity theorem in 11 of 30 joint-study and nine of 25 sequence-study edge-bearing instances. All qualifying instances belong to one identity configuration. Under the theorem's quorum and exporter premises, these witnesses

TABLE II

NATIVE LATE-RELEASE EXECUTIONS. COMPARED DENOTES PREDEFINED SAME-SLICE COMPARISONS; NOSIGNAL IS INCLUDED IN COMPARED. OTHER DENOTES CASES OUTSIDE THE COMPARED AND CROSS-SLICE GROUPS.

Cohort	Runs	Compared	A-first	NoSignal	Cross	Other
Joint-policy	54	36	30	6	16	2
Extension $n = 4$	108	73	59	14	32	3
Extension $n = 7$	24	11	11	0	12	1
Extension total	132	84	70	14	44	4

establish precedence across compatible inputs preserving the required correct-creator facts and comparison domain.

F. Ordering-State and Resource Measurements

In the one-hour runs, MRV evidence updates average $111.14\,\mu$s per export event. Peak aggregate RSS across four primary processes is 323.70 MiB for Tusk and 336.73 MiB for MRV, a 13.03 MiB difference in the observed peaks. The active ordering state is distinct from the longer-lived archive and persistent delivery history. The benchmark-only member-history map reaches 68,093 entries per MRV replica; post-run audits identify 287,552 and 287,924 archive certificates, and 270,020 and 272,372 persistent delivery markers, across the four Tusk and MRV primaries, respectively. Each primary database completes one flush during the observation period, and worker databases exhibit repeated flush and compaction events. The artifact provides the complete resource, processing, and database-event records.

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